

Inventory Management System – Project Statement

The **Inventory Management System** is a desktop-based application developed using **Python, Tkinter, and SQLite3**. The purpose of this project is to provide a simple and efficient system for maintaining product information, managing stock quantities, searching products, and updating inventory records.

The system allows users to register new products, add and update stock quantities, search for products, delete obsolete records, and display the complete inventory. All product and stock information is stored locally in an SQLite database named `inventory.db`, which is automatically created when the application is executed.

The main modules of the system include **Product Registration, Product Search, Stock Update, Product Deletion, and Inventory Display**. The application follows an event-driven programming approach through the Tkinter graphical user interface.

The system requires **Python 3.x, Tkinter, and SQLite3** and can be executed on Windows, Linux, or macOS. Users can run the application using the command `python inventory_system.py`. The database remains saved after closing the application, allowing previously entered information to be accessed when the program is reopened.

The project is designed as a **college-level educational application** for academic demonstration and basic practical inventory management. It is intentionally lightweight and does not include advanced features such as multi-user authentication, cloud synchronization, or enterprise-level security.

Future improvements may include **low-stock alerts, supplier management, sales and purchase invoices, barcode integration, and dashboard-based charts**.

The project consists of the main Python source file `inventory_system.py`, the automatically generated database `inventory.db`, a project report, and a README file containing setup and usage instructions. Before submission, the application should be tested with multiple records and different input conditions, and screenshots of important operations may be included in the project report.